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CONTROLLING TLS VIA ON-CHIP FILTERING TO PREVENT QUBIT ENERGY LOSS

Abstract

Systems and techniques that facilitate controlling TLS via on-chip filtering to prevent qubit energy loss are provided. In various embodiments, a system can comprise a quantum device including a qubit device on a substate. In various embodiments, the quantum device can include an electrode placed in proximity to the qubit device. In various embodiments, an electrical filter can be connected to the electrode. In various embodiments, the quantum device can comprise a voltage source that can be connected to the electrode via the electrical filter. In various embodiments, the voltage source can control a voltage to the electrode to shift a resonant frequency of one or more defects to reduce two level system (TLS) impact on the qubit device.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to controlling two level system (TLS) defects to prevent qubit energy loss within a quantum device to increase performance of related quantum devices, and further, controlling TLS with a voltage supplied via on-chip filtering.

BACKGROUND

[0002] Operations on qubits generally introduce some error in quantum systems, such as some level of decoherence and or some level of quantum noise affecting qubit availability. Such errors can include quantum noise generated by defects called two level systems (TLS). In the present context, the term TLS can refer generally to any material defect that can interact with a qubit and has a characteristic frequency that can be manipulated by an applied electric field. A T1 (energy relaxation time) of a qubit can fluctuate in time due to TLSs. TLSs can cause T1 relaxation times to be shorter than expected, leading to decreases in the coherence times of quantum systems. In addition to TLS, T1 relaxation times can be negatively affected by energy leakage to the environment, an effect called Purcell loss. Decreases in the relaxation and coherence times of quantum systems can limit the ability of the quantum system to perform long quantum algorithms and can increase the error rate of quantum gate operations.

SUMMARY

[0003] The following presents a summary to provide a basic understanding of one or more embodiments of the invention. This summary is not intended to identify key or critical elements, or delineate any scope of the particular embodiments or any scope of the claims. Its sole purpose is to present concepts in a simplified form as a prelude to the more detailed description that is presented later.

[0004] According to an embodiment, a system can comprise a quantum device, an electrode, an electrical filter, and a voltage source. The qubit device can be on a substrate and the electrode can be placed in proximity to the qubit device. Further, the electrical filter can be connected to the electrode, and the voltage source can be connected to the electrode via the electrical filter. The voltage source can control a voltage to the electrode to shift a resonant frequency of the one or more defects to reduce two level system (TLS) impact on the qubit device.

[0005] In one or more embodiments of the aforementioned system, the electrical filter can comprise a filter tuned to reflect signals at a frequency of a transmon qubit of the qubit device. In one or more embodiments of the aforementioned system, the voltage supplied by the voltage source to the electrode can be adjusted to improve the performance of the qubit device. Further, in one or more embodiments, the qubit device can be flux-tunable and the electrical filter can be within λ/4 of the electrode. The electrical filter can be comprised of superconducting materials.

[0006] An advantage of the above-indicated system can be providing for enhanced or recovered qubit performance by applying an electric field to tune the TLS resonant frequency. The presence of the electrode in proximity to the qubit can result in leakage of energy from the qubit to the electrode. In this way, providing a filter near the electrode can be to avoid loss of qubit energy (e.g., Purcell loss) to the voltage source or to the surrounding environment.

[0007] According to another embodiment, a device can comprise a transmon qubit, an electrode, an electrical filter, and a voltage source. The transmon qubit can be on a substrate and the electrode can be placed in proximity to the transmon qubit. Further, the electrical filter can be connected to the electrode, and the voltage source can be connected to the electrode via the electrical filter. The transmon qubit can be disposed on a first surface, and the electrode can be disposed on a second surface.

[0008] In one or more embodiments of the aforementioned device, the first surface and the second surface can be on opposite sides of the substrate. In one or more embodiments of the aforementioned device, the second surface can be on a second substrate. The electrical filter can be disposed on the second surface.

[0009] In one or more embodiments of the aforementioned device, the electrode can be aligned in at least one direction with the transmon qubit. In one or more embodiments of the aforementioned device, the electrical filter can be disposed on a third surface of the second substrate, and wherein the second surface and the third surface can be on opposite sides of the second substrate.

[0010] In one or more embodiments of the aforementioned device, the electrical filter can be within λ/4 of the electrode, and the transmon qubit can be flux-tunable. Additionally, the transmon qubit can be flux-tunable. The electrical filter can comprise a filter tuned to reflect signals at a frequency of the transmon qubit.

[0011] An advantage of the above-indicated device can be providing for enhanced or recovered qubit performance by applying an electric field to tune the TLS resonant frequency. The presence of the electrode in proximity to the qubit can result in leakage of energy from the qubit to the electrode. In this way, providing a filter near the electrode can be to avoid loss of qubit energy (e.g., Purcell loss) to the voltage source or to the surrounding environment.

[0012] According to another embodiment, a method can comprise adjusting a voltage connected to an electrode to shift a resonant frequency of one or more defects to reduce two level system (TLS) impact on the qubit device. The electrode can be located in proximity to the qubit device and a filter can be located between the voltage and the electrode. In embodiments, the filter can be located on a same substrate as the electrode.

[0013] In one or more embodiments of the aforementioned method, the method can comprise measuring a performance of the qubit device. Additionally, the method can comprise adjusting the voltage supplied to the electrode, via a voltage source, to improve the performance of the qubit device. With embodiments, the performance can comprise a Gate Error Rate. In further embodiments, the performance can comprise a T1 time.

[0014] In one or more embodiments of the aforementioned method, the method can comprise improving the performance of the qubit device by: applying, via the voltage source, a sweep of voltage on the electrode over a range via a plurality of small voltage steps; measuring a T1 value of the qubit device as a function of the voltage applied by the voltage source; determining an optimal voltage value that produced a maximum value for the T1 value for the qubit device; and setting the voltage of the electrode to the optimal voltage value.

[0015] In one or more embodiments of the aforementioned method, the method can comprise improving the performance of the qubit device by: applying, via a voltage source, a sweep of voltage on the electrode over a range via a plurality of small voltage steps; measuring a gate error rate of the qubit device as a function of the voltage applied by the voltage source; determining an optimal voltage value that produced a minimum gate error rate for the qubit device; and setting the voltage of the electrode to the optimal voltage value.

[0016] An advantage of the above-indicated method can be providing for enhanced or recovered qubit performance by applying an electric field to tune the TLS resonant frequency. The presence of the electrode in proximity to the qubit can result in leakage of energy from the qubit to the electrode. In this way, providing a filter near the electrode can be to avoid loss of qubit energy (e.g., Purcell loss) to the voltage source or to the surrounding environment.

Description

DESCRIPTION OF THE DRAWINGS

[0017] FIG. **1** illustrates a diagram of an example, non-limiting on-chip filtering system that can facilitate preventing qubit energy loss in accordance with one or more embodiments described

herein.
[0018] FIG. **2** illustrates a diagram of an example, non-limiting on-chip filtering system that can facilitate preventing qubit energy loss in accordance with one or more embodiments described herein.
[0019] FIG. **3** illustrates a diagram of an example, non-limiting on-chip filtering system that can facilitate preventing qubit energy loss in accordance with one or more embodiments described herein.
[0020] FIG. **4** illustrates a block diagram of an example, non-limiting on-chip filtering system that can facilitate preventing qubit energy loss in accordance with one or more embodiments described herein.
[0021] FIG. **5** illustrates a flow diagram of an example, non-limiting method that can facilitate on-chip filtering to prevent qubit energy loss in accordance with one or more embodiments described herein.
[0022] FIG. **6** illustrates a flow diagram of an example, non-limiting method that can facilitate on-chip filtering to prevent qubit energy loss by T1 value in accordance with one or more embodiments described herein.
[0023] FIG. **7** illustrates a flow diagram of an example, non-limiting method that can facilitate on-chip filtering to prevent qubit energy loss by gate error rate in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION

[0024] The following detailed description is merely illustrative and is not intended to limit embodiments and/or application or uses of embodiments. Furthermore, there is no intention to be bound by any expressed or implied information presented in the preceding Background or Summary sections, or in the Detailed Description section.
[0025] One or more embodiments are now described with reference to the drawings, where like referenced numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a more thorough understanding of the one or more embodiments. It is evident, however, in various cases, that the one or more embodiments can be practiced without these specific details.
[0026] As used herein, a quantum circuit can be a set of operations, such as gates, performed on a set of real-world physical qubits with the purpose of obtaining one or more qubit measurements. A quantum processor can comprise the one or more real-world physical qubits.
[0027] Qubit states only can exist (or can only be coherent) for a limited amount of time. Thus, an objective of operation of a quantum logic circuit (e.g., including one or more qubits) can be to maximize the coherence time of the employed qubits. Time spent to operate the quantum logic circuit can undesirably reduce the available time of operation on one or more qubits. This can be due to the available coherence time of the one or more qubits prior to decoherence of the one or more qubits. For example, a qubit state can be lost in less than 100 to 200 microseconds in one or more cases.
[0028] Operation of the quantum circuit can be facilitated, such as by a waveform generator, to produce one or more physical pulses and/or other waveforms, signals and/or frequencies to alter one or more states of one or more of the physical qubits. The altered states can be measured, thus allowing for one or more computations to be performed regarding the qubits and/or the respective altered states.
[0029] Operations on qubits generally can introduce some error, such as some level of decoherence and/or some level of quantum noise, further affecting qubit availability. Quantum noise can refer to noise attributable to the discrete and/or probabilistic natures of quantum interactions.
[0030] A T1 (energy relaxation time) of a qubit can fluctuate in time. One source of the fluctuations can be the noise at the qubit frequency that varies in time. One type of such quantum noise can be due to defects called two level systems (TLS). TLSs are quantum mechanical defects that can exist

in amorphous materials such as glasses or disordered solids. In such amorphous materials, the atoms or molecules are not arranged in a periodic crystal structure but are instead arranged randomly. Such random disorder creates a distribution of potential energy minima and maxima that TLSs can occupy.

[0031] Additionally, the density of TLSs in a material depends on its preparation and history. For instance, the density of TLSs in glasses can be reduced by annealing the glass at high temperatures. However, the annealing process is not always effective and can also create new TLSs. The density of TLSs can also vary depending on the cooling rate and the pressure at which the material was formed.

[0032] Further, TLSs can affect the performance of quantum circuits by introducing fluctuations or local variations in the energy landscape, leading to errors in quantum gate operations. These variations can trap charge carriers, for example, leading to fluctuations in the energy landscape that affect the performance of quantum circuits. The TLS errors can be characterized by the T1, which measures the rate at which quantum information is lost due to relaxation to the ground state.

[0033] TLSs can cause T1 relaxation times to be shorter than expected, leading to a decrease in the coherence time of quantum systems. This decrease in coherence time can limit the ability to perform long quantum algorithms and increase the error rate of quantum gate operations.

[0034] A two-level system has a transition energy (or corresponding frequency). When a TLS is resonant with the qubit frequency, the rate of energy relaxation can increase, leading to shorter T1. TLS frequencies can change as a function of time due to spontaneous changes in the local electric field environment of the TLS. Such changes in the local electric field environment can cause an excellent qubit to suddenly become a poorly functioning qubit (e.g., low T1 times). With examples, an excellent qubit with a T1 time of 500 microseconds can suddenly become a poorly functioning qubit with a T1 time of less than 50 microseconds. With further examples, high performing superconducting qubits can have T1 times greater than 100 microseconds where Purcell loss can affect qubit performance.

[0035] A two-level system (TLS), among other noise causes, can comprise a source of noise that can cause deterioration of coherence parameters (e.g., shorter T1) of one or more qubits of a quantum logic circuit. TLSs are believed to be able to coherently or incoherently couple to the qubit leading to either faster energy relaxation times or rate of energy decay (e.g., shorter T1 s corresponding to an exponential 1/e decay time) as well as faster phase decoherence (e.g., T2). That is, the noise can couple to a low-energy thermal fluctuator, for example, which can randomly change the TLS energy resonance (or the equivalent frequency of the TLS resonance). A TLS can spectrally diffuse into and out of resonance with the qubit frequency when the TLS is in the vicinity of a qubit frequency. This is a source of T1 fluctuation.

[0036] The qubit frequency is the resonance frequency of a qubit energy transition between two states such as, but not limited to, the ground and first excited states of the qubit. The vicinity of a qubit frequency is a frequency range which in some embodiments can range from about 10 megahertz (MHz) below the qubit frequency to about 10 MHz above the qubit frequency. In other embodiments, the vicinity of a qubit frequency can range from about 100 MHz below the qubit frequency to about 100 MHz above the qubit frequency. In still other embodiments, the vicinity of a qubit frequency can range from about 1 gigahertz (GHz) below the qubit frequency to about 1 GHz above the qubit frequency. Without being limited to theory, it is believed that such two-level systems can be caused by atomic scale defects in surface oxides on the metals and/or on the substrate material of a physical real-world qubit and can be electromagnetically active. Indeed, a qubit, such as a transmon itself is a resonator with an electromagnetic excitation, and thus a qubit excitation can couple with a two-level system (TLS) and can cause performance issues for a quantum logic circuit, such as, but not limited to, deterioration of qubit parameters, such as qubit gate error rate.

[0037] Due to presence of two-level systems in/at the quantum system and/or due to maintenance and/or diagnostics to be performed relative to coherence times of a particular qubit, one or more qubits, such as superconducting qubits, can be unavailable and/or not recommended for use with the quantum logic circuit, even if desired for use. Furthermore, absent understanding of such two-level systems and their associated fluctuations relative to the frequency domain of one or more qubits of a quantum system, coherence of the qubit can be affected. Loss of coherence can cause failure of execution of a quantum circuit.

[0038] One or more embodiments are now described with reference to the drawings, wherein like referenced numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a more thorough understanding of the one or more embodiments. It is evident, however, in various cases, that the one or more embodiments can be practiced without these specific details.

[0039] Further, the embodiments depicted in one or more figures described herein are for illustration only, and as such, the architecture of embodiments is not limited to the systems, devices and/or components depicted therein, nor to any particular order, connection and/or coupling of systems, devices and/or components depicted therein. In one or more described embodiments, computer and/or computing-based elements can be used in connection with implementing one or more of the systems, devices, components and/or computer-implemented operations shown and/or described in connection with FIGS. **1-3** and/or with other figures described herein.

[0040] Turning first generally to FIG. **1**, one or more embodiments described herein can include one or more devices, systems and/or apparatuses that can facilitate on-chip filtering to prevent qubit energy loss due to TLS defects. For example, FIG. **1** illustrates a diagram of an example, non-limiting, on-chip filtering system **100** that comprises a quantum device **200** having a qubit device **210**, an electrode **220**, and an electrical filter **230**. The non-limiting, on-chip filtering system **100** to prevent qubit energy loss can additionally include a voltage source **240** connected to the electrode **220** via the electrical filter **230**. The qubit device **210** can be fabricated on a substrate **250** and the electrode **220** can be disposed in proximity (e.g., a distance in which a target qubit can be adjusted without substantially impacting an untargeted qubit) to the qubit device **210**. Further, the voltage source **240** can supply voltage to the electrode **220** to shift a resonant frequency of one or more defects to reduce TLS impact on the qubit device **210**.

[0041] With embodiments, such as generally illustrated in FIG. **1**, the non-limiting, on-chip filtering system **100** to prevent qubit energy loss can apply an electric field via the electrode **220** to recover and/or improve performance of the qubit device **210** (e.g., qubit T1 or qubit gate error rate). TLSs can include resonant frequencies which can coincide with the frequency of the qubit (e.g., a transmon qubit **214** of the qubit device **210**). Further, TLS frequencies can wander/vary in time which can cause rapid reductions in qubit device **210** performance. Such performance can be recovered and/or improved by applying an electric field via the electrode **220**. With examples, the electrode **220** can be disposed over (e.g., in the vertical direction, or in at least one direction) the qubit device **210**. Further, the electrode **220** can provide a pathway for qubit energy to escape, which can be mitigated/blocked by including an electrical filter **230** (e.g., that additionally blocks outside energy from entering the qubit device **210**). However, placing an electrode (e.g., the electrode **220**) in proximity to transmon qubit **214** may result in energy leakage, such as Purcell loss, from the qubit.

[0042] In examples, the electrode **220** can be one or more of a variety of components that can generate/produce an electric field proximate a qubit device **210** to effectively tune the TLS defect frequency away from the qubit device **210** frequency. In an embodiment, the electrode **220** can be one or more loops of one or more wires, or one or more lines, whereby a common voltage can be applied to tune the frequency of the TLS defects, and such loops, wires, or lines may additionally be used in flux tuning of quantum components. In some embodiments, the electrode **220** can be a thin film of superconducting metal that is disposed in close proximity to the qubit device **210**. For example and without limitation, the electrode **220** can be disposed apart from the qubit device **210** by about 50 micrometers or more or less. In examples, with an increase in the distance between the electrode **220** and the qubit device **210**, the voltage can be respectively increased by the voltage source **240** to compensate for any decrease in the electric field generated by the electric voltage.

[0043] In embodiments, the electrical filter **230** can provide one or more of a variety of functions for the qubit device **210** and/or the on-chip filtering system **100**. The electrical filter **230** can reflect qubit energy (e.g., energy lost through Purcell loss) back into the qubit device **210**, and the electrical filter **230** can block outside energy/noise from entering and interfering with performance of the qubit device **210**. Additionally, by placing the electrical filter **230** near (e.g., within a suitable distance to mitigate loss, such as λ/4 of the qubit device **210**) the electrode **220**, energy loss that occurs during signal propagation may be limited. In this way, performance of the qubit device **210** can be enhanced/improved by including an electrode **220** and an electrical filter **230** coupled with a voltage source **240** to provide a controlled electric field thereby controlling the frequency of one or more TLSs of the qubit device **210** and blocking external interferences and energy loss from the transmon qubit **214** itself. TLS defects include two energy levels where the difference in energy can determine the characteristic resonant frequency of the TLS. The TLS frequency can depend on the atomic configuration of the qubit materials and the presence of an electric field. TLS defects include an associated electric dipole moment that can allow for frequency tuning via an applied electric field. The electrode **220** can generate an electrical field via the voltage source **240** such that the frequency of the TLS defect can be tuned (e.g., tuned away from the qubit device **210** frequency).

[0044] Repetitive description of like elements employed in other embodiments described herein is omitted for sake of brevity. Aspects of systems (e.g., the on-chip filtering system **100** to prevent qubit energy loss and the like), apparatuses or processes in various embodiments of the present invention can constitute one or more machine-executable components embodied within one or more machines (e.g., embodied in one or more computer readable mediums (or media) associated with one or more machines). Such components, when executed by the one or more machines (e.g., computers, computing devices, virtual machines, a combination thereof, and/or the like) can cause the machines to perform the operations described.

[0045] As illustrated in FIG. **1**, the non-limiting, on-chip filtering system **100** to prevent qubit energy loss can include a first substrate **250** and a second substrate **252**. The first substrate **250** can be disposed substantially parallel to the second substrate **252**. The first substrate **250** can be one or more of a variety of qubit chips and can include the qubit device **210**. Further, the qubit device **210** can be disposed on the first substrate **250**. The first substrate **250** can include a first surface **270** and a second surface **272**, where the first surface **270** can be opposite the second surface **272**. The qubit device **210** can be disposed on the first surface **270** of the first substrate **250**. The electrode **220** and the electrical filter **230** can be disposed on the second substrate **252** (see, e.g., FIG. **1**). Further, the second substrate **252** can include a third surface **274** opposite a fourth surface **276**. The second substrate **252** can be one or more of a variety of interposer chips; and the second substrate **252** can include ancillary circuitry for the on-chip filtering system **100** or for a variety of other connected systems.

[0046] Such as illustrated in FIG. **1**, the on-chip filtering system **100** to prevent qubit energy loss can include an additional qubit device **212**, an additional electrode **222**, an additional electrical filter **232**, and an additional voltage source **242**. With embodiments, any number of qubit devices **210** can be included on the first substrate **250** and/or the second substrate **252**.

[0047] In embodiments, the qubit device **210** and the additional qubit device **212** can include a transmon qubit **214** and an additional transmon qubit **216**, respectively. The transmon qubit **214** and the additional transmon qubit **216** can be one or more of a variety of qubits. For example and without limitation, the transmon qubit **214** and the additional transmon qubit **216** can be flux-tunable qubits and/or phase qubits. The electrical filter **230** and the additional electrical filter **232** can include filters that can be tuned to reflect signals at a frequency of a transmon qubit **214** and the additional transmon qubit **216** (e.g., the qubit device **210** and the additional qubit device **212**).

[0048] With embodiments, such as generally illustrated in FIG. **1**, the non-limiting on-chip filtering system can include the voltage source **240** and the additional voltage source **242** coupled with the electrical filter **230** and the additional electrical filter **232**, respectively. The voltage source **240** and the additional voltage source **242** can supply power/voltage to the electrode **220** and the additional electrode **222**, via the electrical filter **230** and the additional electrical filter **232**, respectively. Supplying voltage to the electrode **220** and the additional electrode **222** can shift the resonant frequency of one or more defects of the non-limiting, on-chip filtering system **100** to reduce TLS impact (e.g., interference and/or noise) on the qubit device **210** and the additional qubit device **212**. With examples, the voltage source **240** and the additional voltage source **242** can supply a variety of voltages ranging from about −10V to about 10V (e.g., in small steps of about 10 mV or more or less). Additionally, the electrical filter **230** and the additional electrical filter **232** can reflect signals at the frequency of the qubit device **210** (e.g., the transmon qubit **214**) and the additional qubit device **212** (e.g., the additional transmon qubit **216**) such that outside noise originating from an environment exterior to the on-chip filtering system **100** does not affect performance of the transmon qubit **214** or the additional transmon qubit **216**.

[0049] In examples, the first substrate **250** can be a qubit chip, whereby the first substrate **250** can include the qubit device **210** and the additional qubit device **212**. The second substrate **252** can be an interposer chip that can include various circuity for supporting/controlling the qubit device **210** and the additional qubit device **212**. Further, as indicated in FIG. **1**, the first substrate **250** (e.g., the qubit chip) can be bump-bonded with the second substrate **252** (e.g., the interposer chip) via one or more bump-bonds **260**, **262**. The first substrate **250** can be bump-bonded with the second substrate **252** such that the first surface **270** of the first substrate **250** is proximate the third surface **274** of the second substrate **252**. Further, the electrode **220** and the additional electrode **222** disposed on the second substrate **252** can be in close proximity to the qubit device **210** and the additional qubit device **212** disposed on the first substrate **250**. With examples, the electrode **220** and the additional electrode **222** can be aligned (e.g., at least partially, or fully, overlapping) in at least one direction (e.g., vertically) with the qubit device **210** and the additional qubit device **212**. The electrode **220** and the additional electrode **222** can be aligned in a first direction (e.g., a horizontal direction along the second substrate **252**) with the electrical filter **230** and the additional electrical filter **232**; and the electrode **220** and the additional electrode **222** can be aligned in a second direction (e.g., a vertical direction perpendicular to the first direction) with the transmon qubit **214** and the additional transmon qubit **216**.

[0050] Turning next to FIG. **2**, the non-limiting, on-chip filtering system **100** to prevent qubit energy loss can include one or more of a variety of configurations. For example and without limitation, the electrical filter **230** and the additional electrical filter **232** can be disposed on a different surface than the electrodes **220**, **222** and the qubit devices **210**, **212**. The electrical filter **230** and the additional electrical filter **232** can be disposed on the fourth surface **276** of the second substrate **252**. The electrical filter **230** and the additional electrical filter **232** can be disposed on an opposite surface with respect to the electrode **220** and the additional electrode **222**; further, the electrical filter **230** and the additional electrical filter **232** can be disposed on opposite sides of the second substrate **252**.

[0051] With embodiments, the non-limiting, on-chip filtering system **100** to prevent qubit energy loss can include one or more thru substrate vias (TSVs) that can couple the electrical filter **230** to the electrode **220** and can couple the additional electrical filter **232** to the additional electrode **222**, such that voltage can be applied through the second substrate **252**. In examples, the electrical filter **230** and the additional electrical filter **232** can be disposed on an opposite side with respect to the electrode **220** and the additional electrode **222** to preserve wiring space and/or minimize interferences with the on-chip filtering system **100**. The on-chip filtering system **100** can include a TSV **280** and an additional TSV **282**.

[0052] In embodiments, such as generally illustrated in FIG. **2**, the TSV **280** can electrically couple the voltage source **240** and the electrical filter **230** with the electrode **220** such that the voltage source **240** can supply a voltage to the electrode **220** to create an electric field within the vicinity (e.g., proximate) of the qubit device **210** to change the frequencies of the TLSs such as to not interfere with qubit device **210** performance (e.g., qubit T1). The additional TSV **282** can electrically couple the additional voltage source **242** and the additional electrical filter **232** with the additional electrode **222** such that the additional voltage source **242** can supply a voltage to the additional electrode **222** to create an electric field within the vicinity (e.g., proximate) of the additional qubit device **212** to change the frequencies of the TLSs such as to not interfere with additional qubit device **212** performance (e.g., qubit T1).
[0053] Turning to FIG. **3**, the non-limiting, on-chip filtering system **100** to prevent qubit energy loss can include a configuration where the qubit device **210**, the electrode **220**, and the electrical filter **230** can be disposed on the same substrate (e.g., the first substrate **250**). Alternatively, in one or more various embodiments, the electrical filter **230** can be disposed on the second substrate **252**. The qubit device **210** can be disposed on the first surface **270** of the first substrate **250**; and the electrode **220** and the electrical filter **230** can be disposed on the second surface **272** of the first substrate **250**. In a similar manner, the additional qubit device **212**, the additional electrode **222**, and the additional electrical filter **232** can be disposed on the same substrate (e.g., the first substrate **250**). Alternatively, in one or more various embodiments, the additional electrical filter **232** can be disposed on the second substrate **252**. The additional qubit device **212** can be disposed on the first surface **270** of the first substrate **250**; and the additional electrode **222** and the electrical filter **232** can be disposed on the second surface **272** of the first substrate **250**.
[0054] With embodiments, such as generally illustrated in FIGS. **1**-**3**, the electrode **220** and the additional electrode **222** can be disposed less than about a quarter of a wavelength (λ/4) away from the electrical filter **230** and the additional electrical filter **232**, respectively. The wavelength A can be determined by the frequency f of the qubit and the propagation velocity v of signals in the wiring according to λ=v/f. In examples, a quarter of a wavelength can be less than about 1 centimeter.
[0055] In embodiments, such as generally illustrated in FIG. **4**, the on-chip filtering system **100** to prevent qubit energy loss can include an electrical filter **230**. The electrical filter **230** can include one or more of a variety of components and/or types. For example and without limitation, the electrical filter **230** can be any type/arrangement of filter elements to result in a low-pass response with a cut-off frequency of about 1.5 GHz to about 2.0 GHz or more or less. That is, the frequency of the qubit device **210** (e.g., the transmon qubit **214**) can be about 5 GHz, and the electrical filter **230** can be connected/selected to reject and reflect signals with frequencies higher than the filter cut-off frequencies, including the frequency of the qubit device **210**.
[0056] With examples, the electrical filter **230** can be any variety of low loss filter and can be constructed from superconducting elements. Further, the electrical filter **230** can be a low pass filter, a band reject filter (e.g., consisting of an open-ended quarter-wave stub fabricated using coplanar waveguide circuity), and a combination of low pass and band reject filters (e.g., for a more complete rejection of energy leakage from the qubit and to protect the introduction of noise at both the qubit frequency and the resonator readout frequency.
[0057] In examples, the qubit device **210** can be a transmon qubit **214** and can further comprise a first capacitor plate **218**A, a second capacitor plate **218**B, and a Josephson junction **219**, as shown in FIG. **4**. The Josephson junction **219** can bridge the first capacitor plate **218**A with the second capacitor plate **218**B.
[0058] As illustrated in FIG. **4**, the electrode **220** can be coupled (e.g., electrically) with the electrical filter **230** which can consist of an inductor **230**A and a capacitor **230**B. The electrical filter **230** can reject frequencies that are similar/near to the frequency of the qubit device **210** (e.g., frequencies greater than about 1.5 GHz). The electrical filter **230** can effectively prevent/limit

energy of the qubit device **210** from leaving and can prevent noise from entering and disturbing the qubit device **210**. With embodiments, the inductor **230**A and the capacitor **230**B can be one or more of a variety of inductors or capacitors used with quantum devices. For example, the inductor **230**A can be a spiral inductor and the capacitor **230**B can be implemented as a planar interdigitated capacitor. The filter components can be made from superconducting materials (e.g., niobium, aluminum, tantalum, titanium nitride, etc.) in order to avoid electrical dissipation. Additionally, the input of the electrical filter **230** can be coupled to external circuitry **290**, which can include circuit boards, coaxial cables, and additional filters for the on-chip filtering system **100**. The external circuitry **290** can be coupled to and/or can include the voltage source **240** that provides a voltage to the electrode **220** for tuning the frequencies of the TLSs.

[0059] In embodiments, the voltage source **240** can supply a voltage that can range from about −10V to about 10V to be applied to the electrode **220**. The electrode **220** (e.g., the voltage source **240**) can generate an electric field proximate the qubit device **210** that can change the frequencies of the TLSs such that an optimal voltage (V.sub.opt) can be found to improve performance of the qubit device **210**. The non-limiting, on-chip filtering system **100** to prevent qubit energy loss can adjust the voltage supplied to the electrode **220**, via the voltage source **240**, to improve performance of the qubit device **210** (e.g., to improve T1 of the qubit device **210**) as TLS defect frequencies change/vary during operation of the non-limiting, on-chip filtering system **100** and the qubit device **210**. The voltage source **240** can provide a sweep of voltages such that the qubit device **210** produces maximum T1 values (e.g., where qubit coherence time is improved) and can maintain performance of the qubit device **210** above a minimum performance value (e.g., T1.sub.setpoint). Alternatively, the voltage supplied by the voltage source **240** can be chosen to minimize the gate error rate of quantum gate operations.

[0060] FIG. **5** illustrates a flow diagram of an example, non-limiting method **500** to facilitate on-chip filtering to prevent qubit energy loss in accordance with one or more embodiments described herein. Performance of the on-chip filtering system **100** can be improved with respect to measurement of the qubit relaxation time T1, where a minimum acceptable T1 value (e.g., T1 setpoint) can be set and the non-limiting method **500** can maintain qubit performance above the minimum acceptable T1 value. Repetitive description of like elements and/or processes employed in respective embodiments is omitted for sake of brevity.

[0061] At **502**, the non-limiting method **500** to facilitate on-chip filtering to prevent qubit energy loss can comprise adjusting a voltage connected to an electrode **220** to shift a resonant frequency of one or more defects to reduce TLS impact on the qubit device **210**.

[0062] At **504**, the non-limiting method **500** to facilitate on-chip filtering to prevent qubit energy loss can comprise measuring a performance of the qubit device **210**. In examples, performance can comprise a gate error rate and/or a T1 time.

[0063] At **506**, the non-limiting method **500** to facilitate on-chip filtering to prevent qubit energy loss can comprise adjusting the voltage supplied to the electrode **220**, via a voltage source **240**, to improve performance of the qubit device **210**.

[0064] Additionally, step **506** can further be described by the continued flowchart illustrated in FIG. **6**. Adjusting the voltage supplied to the electrode **220** to improve performance of the qubit device **210** can comprise one or more of a variety of steps. For examples, at **520**, the non-limiting method **500** can comprise selecting a minimum acceptable T1 value (e.g., T1.sub.setpoint).

[0065] At **522**, the non-limiting method **500** can comprise applying a sweep of voltages on the electrode **220** over a range (e.g., a range of −10V to +10V) via small voltage steps from the voltage source **240** (e.g., voltage steps of about 10 mV or more or less).

[0066] At **524**, the non-limiting method **500** can comprise measuring the T1 value of the qubit device **210** as a function of the applied voltage from the voltage source **240** (e.g., T1(V)).

[0067] At **526**, the non-limiting method **500** can comprise determining if the data (e.g., the T1(V)) is noisy. In response a determination of noisy data, at **528**, the non-limiting method **500** can

comprise smoothing the data by averaging adjacent T1 values over the data (e.g., T1(V)).
[0068] At **526**, the non-limiting method **500** can determine the data is not noisy and can proceed to step **530** without performing step **528**. Further, at **530**, the non-limiting method **500** can comprise determining the optimal voltage value (V.sub.Opt) that produced the maximum value of T1 based on the data (e.g., T1(V)).
[0069] At **532**, the non-limiting method **500** can comprise setting the voltage of the electrode **220** to the optimal voltage value (V.sub.Opt).
[0070] At **534**, the non-limiting method **500** can comprise periodically checking the T1 value. Additionally, at **526**, the non-limiting method **500** can comprise determining if the T1 value is less than the minimum acceptable T1 value (e.g., T1.sub.setpoint). In response to a determination that the T1 value is not less than the T1 setpoint, the non-limiting method **500** can perform step **522** of applying a sweep of voltages to determine the optimal voltage value (V.sub.Opt). If the T1 value is greater than the minimum acceptable T1 value, the non-limiting method **500** can continue to periodically check the T1 value to verify performance of the qubit device **210**.
[0071] Additionally or alternatively, step **506** can further be described by the continued flowchart illustrated in FIG. **7**. Adjusting the voltage supplied to the electrode **220** to improve performance of the qubit device **210** can comprise one or more of a variety of steps. For example, the variety of steps can involve evaluating a Gate Error Rate (GER) of the qubit device **210** (e.g., in contrast to evaluating performance based on T1 values). At **540**, the method **500** can comprise selecting a maximum acceptable GER of the qubit device **210**.
[0072] At **542**, the non-limiting method **500** can comprise applying a sweep of voltages on the electrode **220** over a range (e.g., a range of −10V to +10V) via small voltage steps from the voltage source **240** (e.g., voltage steps of about 10 mV or more or less).
[0073] At **544**, the non-limiting method **500** can comprise measuring the GER of the qubit device **210** as a function of the applied voltage from the voltage source **240** (e.g., GER(V)).
[0074] At **546**, the non-limiting method **500** can comprise determining if the data (e.g., the GER(V)) is noisy. In response a determination of noisy data, at **548**, the non-limiting method **500** can comprise smoothing the data by averaging adjacent GER values over the data (e.g., GER(V)).
[0075] At **546**, the non-limiting method **500** can determine the data is not noisy and can proceed to step **550** without performing step **548**. Further, at **550**, the non-limiting method **500** can comprise determining the optimal voltage value (V.sub.Opt) that produced the minimum value of GER based on the data (e.g., GER(V)).
[0076] At **552**, the non-limiting method **500** can comprise setting the voltage of the electrode **220** to the optimal voltage value (V.sub.Opt).
[0077] At **554**, the non-limiting method **500** can comprise periodically checking the GER value. Additionally, at **556**, the non-limiting method **500** can comprise determining if the GER value is less than the maximum acceptable GER value. In response to a determination that the GER value is greater than the maximum acceptable GER, the non-limiting method **500** can perform step **542** of applying a sweep of voltages to determine the optimal voltage value (V.sub.Opt). If the GER value is less than the maximum acceptable GER value, the non-limiting method **500** can continue to periodically check the GER value to verify performance of the qubit device **210**.
[0078] Aspects of the one or more embodiments described herein are described herein with reference to flowchart illustrations or block diagrams of methods, apparatus (systems), and devices according to one or more embodiments described herein. It will be understood that each block of the flowchart illustrations or block diagrams, and combinations of blocks in the flowchart illustrations or block diagrams, can be implemented by computer readable program instructions. These computer readable program instructions can be provided to a processor of a general purpose computer, special purpose computer or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts

specified in the flowchart or block diagram block or blocks. These computer readable program instructions can also be stored in a computer readable storage medium that can direct a computer, a programmable data processing apparatus or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart or block diagram block or blocks. The computer readable program instructions can also be loaded onto a computer, other programmable data processing apparatus or other device to cause a series of operational acts to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus or other device implement the functions/acts specified in the flowchart or block diagram block or blocks.

[0079] The flowcharts and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, computer-implementable methods or computer program products according to one or more embodiments described herein. In this regard, each block in the flowchart or block diagrams can represent a module, segment or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s). In one or more alternative implementations, the functions noted in the blocks can occur out of the order noted in the Figures. For example, two blocks shown in succession can, in fact, be executed substantially concurrently, or the blocks can sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams or flowchart illustration, and combinations of blocks in the block diagrams or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

[0080] While the subject matter has been described above in the general context of computer-executable instructions of a computer program product that runs on a computer or computers, those skilled in the art will recognize that the one or more embodiments herein also can be implemented in combination with other program modules. Generally, program modules include routines, programs, components, data structures or the like that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the inventive computer-implemented methods can be practiced with other computer system configurations, including single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as computers, hand-held computing devices (e.g., PDA, phone), microprocessor-based or programmable consumer or industrial electronics or the like. The illustrated aspects can also be practiced in distributed computing environments in which tasks are performed by remote processing devices that are linked through a communications network. However, some, if not all aspects of the one or more embodiments can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0081] As used in this application, the terms “component,” “system,” “platform,” “interface,” or the like, can refer to or can include a computer-related entity or an entity related to an operational machine with one or more specific functionalities. The entities disclosed herein can be either hardware, a combination of hardware and software, software, or software in execution. For example, a component can be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program or a computer. By way of illustration, both an application running on a server and the server can be a component. One or more components can reside within a process or thread of execution and a component can be localized on one computer or distributed between two or more computers. In another example, respective components can execute from various computer readable media having various data structures stored thereon. The components can communicate via local or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor. In such a case, the processor can be internal or external to the apparatus and can execute at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, where the electronic components can include a processor or other means to execute software or firmware that confers at least in part the functionality of the electronic components. In an aspect, a component can emulate an electronic component via a virtual machine, e.g., within a cloud computing system.

[0082] In addition, the term "or" is intended to mean an inclusive "or" rather than an exclusive "or." That is, unless specified otherwise, or clear from context, "X employs A or B" is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then "X employs A or B" is satisfied under any of the foregoing instances. Moreover, articles "a" and "an" as used in the subject specification and annexed drawings should generally be construed to mean "one or more" unless specified otherwise or clear from context to be directed to a singular form. As used herein, the terms "example" and/or "exemplary" are utilized to mean serving as an example, instance, or illustration. For the avoidance of doubt, the subject matter disclosed herein is not limited by such examples. In addition, any aspect or design described herein as an "example" and/or "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects or designs, nor is it meant to preclude equivalent exemplary structures and techniques known to those of ordinary skill in the art.

[0083] As it is employed in the subject specification, the term "processor" can refer to substantially any computing processing unit or device comprising, but not limited to, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. Further, processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

[0084] Herein, terms such as "store," "storage," "data store," data storage," "database," and substantially any other information storage component relevant to operation and functionality of a component are utilized to refer to "memory components," entities embodied in a "memory," or components comprising a memory. It is to be appreciated that memory or memory components described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory. By way of illustration, and not limitation, nonvolatile memory can include read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), flash memory or nonvolatile random access memory (RAM) (e.g., ferroelectric RAM (FeRAM). Volatile memory can include RAM, which can act as external cache memory, for example. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), direct Rambus RAM (DRRAM), direct Rambus

dynamic RAM (DRDRAM) or Rambus dynamic RAM (RDRAM). Additionally, the disclosed memory components of systems or computer-implemented methods herein are intended to include, without being limited to including, these and any other suitable types of memory.
[0085] What has been described above include mere examples of systems and computer-implemented methods. It is, of course, not possible to describe every conceivable combination of components or computer-implemented methods for purposes of describing the one or more embodiments, but one of ordinary skill in the art can recognize that many further combinations and permutations of the one or more embodiments are possible. Furthermore, to the extent that the terms “includes,” “has,” “possesses,” and the like are used in the detailed description, claims, appendices and drawings such terms are intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.
[0086] The descriptions of the one or more embodiments provided herein have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A system comprising: a quantum device comprising: a qubit device on a substrate; an electrode placed in proximity to the qubit device; an electrical filter connected to the electrode; and a voltage source connected to the electrode via the electrical filter, wherein the voltage source controls a voltage to the electrode to shift a resonant frequency of one or more defects to reduce two level system (TLS) impact on the qubit device.
2. The system of claim 1, wherein the electrical filter comprises a filter tuned to reflect signals at a frequency of a transmon qubit of the qubit device.
3. The system of claim 1, wherein the voltage supplied by the voltage source to the electrode is adjusted to improve performance of the qubit device.
4. The system of claim 1, wherein the qubit device is flux-tunable.
5. The system of claim 1, wherein the electrical filter is within **24** of the electrode.
6. The system of claim 1, wherein the electrical filter is comprised of superconducting materials.
7. A device comprising: a transmon qubit on a substrate; an electrode placed in proximity to the transmon qubit; an electrical filter connected to the electrode; and a voltage source connected to the electrode via the electrical filter, wherein the transmon qubit is disposed on a first surface, and the electrode is disposed on a second surface.
8. The device of claim 7, wherein the first surface and the second surface are on opposite sides of the substrate.
9. The device of claim 7, wherein the second surface is on a second substrate.
10. The device of claim 7, wherein the electrical filter is disposed on the second surface.
11. The device of claim 7, wherein the electrode is aligned in at least one direction with the transmon qubit.
12. The device of claim 9, wherein the electrical filter is disposed on a third surface of the second substrate, and wherein the second surface and the third surface are on opposite sides of the second substrate.
13. The device of claim 7, wherein the electrical filter is within $\lambda/4$ of the electrode.
14. The device of claim 7, wherein the transmon qubit is flux-tunable.
15. The device of claim 7, wherein the electrical filter comprises a filter tuned to reflect signals at a frequency of the transmon qubit.

16. A method comprising: adjusting a voltage connected to an electrode to shift a resonant frequency of one or more defects to reduce two level system (TLS) impact on a qubit device, wherein the electrode is located in proximity to the qubit device and a filter is located between the voltage and the electrode, and wherein the filter is located on a same substrate as the electrode.
17. The method of claim 16, further comprising: measuring a performance of the qubit device; and adjusting the voltage supplied to the electrode, via a voltage source, to improve the performance of the qubit device.
18. The method of claim 17, wherein the performance comprises a Gate Error Rate.
19. The method of claim 17, wherein the performance comprises a T1 time.
20. The method of claim 17, further comprising: improving the performance of the qubit device by: applying, via the voltage source, a sweep of voltage on the electrode over a range via a plurality of small voltage steps; measuring a T1 value of the qubit device as a function of the voltage applied by the voltage source; determining an optimal voltage value that produced a maximum value for the T1 value for the qubit device; and setting the voltage of the electrode to the optimal voltage value.
21. The method of claim 17, further comprising: improving the performance of the qubit device by: applying, via the voltage source, a sweep of voltage on the electrode over a range via a plurality of small voltage steps; measuring a gate error rate of the qubit device as a function of the voltage applied by the voltage source; determining an optimal voltage value that produced a minimum gate error rate for the qubit device; and setting the voltage of the electrode to the optimal voltage value.